Automated Service Case Management using Salesforce, OCR & AI

A Project Work-I Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

BY

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EN22IT301072

Under the Guidance of
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Department of Information Technology
Faculty of Engineering
MEDICAPS UNIVERSITY, INDORE- 453331

JAN-JUN 2025-2026

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Report Approval

The project work **“AUTOMATED SERVICE CASE MANAGEMENT USING SALESFORCE,OCR AND AI”** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

I hereby declare that the project entitled **"AUTOMATED SERVICE CASE MANAGEMENT USING SALESFORCE,OCR AND AI"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in 'Information Technology' completed under the supervision of Prof Rahul Singh Pawar**, designation, department of Information Technology,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Priyal Bahety

EN22IT301072

Certificate

I, Prof Rahul Singh Pawar certify that the project entitled **“AUTOMATED SERVICE CASE MANAGEMENT USING SALESFORCE,OCR AND AI”** submitted in partial fulfillment for the award of the degree of Bachelor of Technology by Priyal Bahety is the record carried out by her under my guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

The project Automated Service Case Management using Salesforce, OCR & AI is developed to automate customer service request handling for appliances such as washing machines, refrigerators, televisions, and air conditioners. In many organizations, customer complaints are handled manually through emails, causing delays, errors, and inefficient routing of service cases.

This system integrates Salesforce CRM, Optical Character Recognition (OCR), and Artificial Intelligence (AI) to streamline the complete process. When a customer sends an email with attachments such as invoices, warranty cards, or product images, Salesforce automatically creates a service case. OCR technology extracts useful text and data from images and PDF files. AI then analyses the extracted content and classifies the case according to issue type, urgency, and department requirement.

The system reduces manual work, improves response speed, increases accuracy in case classification, and ensures better customer satisfaction. It also helps organizations manage large numbers of service requests efficiently.

This project demonstrates how intelligent automation can improve customer support operations and provide faster, smarter, and more reliable service management solutions.

Keywords:

Salesforce, OCR, AI, Automation, Customer Service, Case Management

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Abbreviations

S.No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	OCR	Optical Character Recognition
3	CRM	Customer Relationship Management
4	PDF	Portable Document Format
5	UI	User Interface
6	API	Application Programming Interface
7	DB	Database
8	ML	Machine Learning
9	IT	Information Technology
10	OS	Operating System
11	CPU	Central Processing Unit
12	RAM	Random Access Memory
13	KPI	Key Performance Indicator
14	SLA	Service Level Agreement
15	URL	Uniform Resource Locator

Notations & Symbols

S.No.	Symbol / Notation	Meaning
1	→	Process Flow / Direction
2	↓	Next Step
3	+	Addition / Combined Process
4	%	Percentage
5	#	Number
6	*	Important / Mandatory
7	@	Email/User Reference
8	=	Equal To
9	>	Greater Than
10	<	Less Than
11	₹	Indian Rupee
12	$	Dollar Currency
13	✓	Success / Completed
14	✗	Failure / Error
15	:	Description / Ratio

Chapter-1:Introduction

1.1 Introduction

Customer service management is one of the most important functions in any organization. Companies that sell appliances such as washing machines, refrigerators, televisions, and air conditioners receive many customer complaints, service requests, and product inquiries every day. Customers usually expect quick responses and fast solutions to their problems. However, in many organizations, customer service requests are still handled manually through emails, phone calls, and paper-based systems.

Manual handling of customer cases creates several challenges such as delays in response, human errors, incorrect case routing, and high workload on support teams. Employees need to read emails, check attachments, create service cases manually, assign priorities, and forward them to the concerned department. This process consumes time and affects customer satisfaction.

To solve these issues, this project introduces an **Automated Service Case Management System using Salesforce, OCR, and AI**. The system uses Salesforce CRM to automatically create service cases from emails. OCR technology extracts important text from images and PDF documents such as invoices, warranty cards, and complaint images. Artificial Intelligence helps classify the case type, set priority level, and route the case to the correct team.

This automation reduces manual effort, improves efficiency, increases speed and accuracy, and provides better customer support services.

```
Customer Sends Email
        ↓
Support Staff Reads Email
        ↓
Checks Attachments Manually
        ↓
Creates Case Manually
        ↓
Assigns Priority Manually
        ↓
Routes to Team
        ↓
Delayed Resolution
```

Figure 1.1 Existing Manual Case Management System

1.2 Objectives

The main objectives of this project are:

- To automate service case creation from customer emails.
- To extract useful information from images and PDF files using OCR.
- To classify service requests using Artificial Intelligence.
- To assign case priority automatically.
- To reduce manual workload of support staff.
- To improve customer response time and satisfaction.
- To increase overall efficiency of customer service operations.

```
Customer Email
     ↓
Salesforce Email-to-Case
     ↓
OCR Extracts Data
     ↓
AI Classifies Issue
     ↓
Priority Assigned
     ↓
Auto Routing
     ↓
Quick Resolution
```

Figure 1.2 Proposed Automated Service Case Management Overview

---

1.3 Scope of the Project

The scope of this project is wide and useful for many industries. It can be implemented in:

- Home appliance service companies
- Electronics repair centers
- Customer support departments
- CRM-based organizations

- Warranty claim processing systems
- Complaint management systems

The project can also be extended in future with chatbot support, voice-based complaints, and mobile applications.

1.4 Need of the Project

There is a growing need for automation in customer service because manual processes are slow and inefficient. Companies receive a large number of complaints daily, making it difficult to manage cases manually. Delays in solving customer issues reduce trust and satisfaction. Therefore, an intelligent automated system is needed to handle customer requests quickly and accurately.

1.5 Advantages of the Project

- Faster case creation and handling
- Improved accuracy in classification
- Automatic extraction of data from documents
- Better customer experience
- Reduced operational cost
- Less manual effort
- Better tracking and monitoring of service cases

1.6 Organization of Report

This report is organized into different chapters. Chapter 1 explains the introduction and objectives. Chapter 2 discusses system requirements and analysis. Chapter 3 explains system working and flow. Chapter 4 covers design and architecture. Chapter 5 describes testing. Chapter 6 presents limitations. Chapter 7 discusses future scope. Chapter 8 gives conclusion and Chapter 9 provides references.

CHAPTER 2: SYSTEM REQUIREMENT ANALYSIS

2.1 Introduction

System Requirement Analysis is an important phase in software development. It helps in understanding the needs of the system, identifying resources, and planning the implementation process. For the project **Automated Service Case Management using Salesforce, OCR & AI**, requirement analysis is necessary to ensure smooth automation of customer service case handling.

This chapter explains the existing system, problems faced in manual operations, proposed automated solution, and hardware/software requirements needed for successful implementation.

S.No.	Component	Specification
1	Processor	Intel Core i3 or Above
2	RAM	Minimum 4 GB
3	Hard Disk	250 GB or Above
4	Monitor	15 Inch or Above
5	Internet	Broadband Connection

Table 2.1 Hardware Requirements

```
Requirements
    ├── Functional
    │       ├── Create Cases
    │       ├── OCR Extraction
    │       ├── AI Classification
    │       └── Auto Routing
    │
    └── Non Functional
            ├── Security
            ├── Accuracy
            ├── Speed
            └── Scalability
```

Fig.2.1 Requirement Analysis Diagram

2.2 Existing System

In many organizations, customer complaints and service requests are managed manually. Customers usually send emails regarding issues such as product damage, leakage, warranty claims, invoice requests, or service scheduling. Employees then read emails manually and create cases in the system.

The manual process generally includes:

- Reading customer emails one by one
- Downloading attachments manually
- Checking invoices or warranty cards
- Creating service tickets manually
- Deciding priority level manually
- Forwarding cases to departments manually
- Tracking updates manually

This traditional method consumes time and resources.

S.No.	Software	Purpose
1	Windows 10/11	Operating System
2	Salesforce CRM	Case Management
3	OCR Tool	Text Extraction
4	OpenAI / AI Tool	Case Classification
5	Google Chrome	Browser
6	MS Word	Documentation

Table 2.2 Software Requirements

```
Existing System             Proposed System
--------------             ----------------
Manual Entry               Auto Case Creation
Slow Response              Fast Response
Human Errors               AI Accuracy
Manual Routing             Smart Routing
High Workload              Less Workload
```

Fig 2.2 Existing Vs Proposed System Comparison

2.3 Problems in Existing System

The manual system creates several challenges:

- Delay in case creation and response
- Human errors in data entry
- Wrong case classification
- High workload on employees
- Poor customer satisfaction
- Difficulty in handling large number of requests
- Lack of automation and smart routing
- Slow decision making

These problems reduce the overall efficiency of customer service operations.

2.4 Proposed System

To overcome the limitations of the existing system, an automated solution is proposed using Salesforce, OCR, and Artificial Intelligence.

In the proposed system:

- Customer emails automatically generate service cases through Salesforce Email-to-Case.
- OCR extracts text from invoices, PDFs, and images.
- AI classifies cases based on issue type and urgency.
- Priority is assigned automatically.
- Cases are routed to the correct department.
- Status can be tracked easily in Salesforce CRM.

This system improves speed, accuracy, and efficiency.

2.5 Functional Requirements

The system should provide the following functions:

- Automatic case creation from emails
- Upload and read attachments
- OCR text extraction from images/PDFs

- AI-based case classification
- Automatic priority assignment
- Case routing to support teams
- Status tracking and reporting
- User login and access control

2.6 Non-Functional Requirements

- Fast response time
- Secure data handling
- High accuracy in classification
- Easy user interface
- Reliable system performance
- Scalable for large number of cases
- Easy maintenance

2.7 Hardware Requirements

Minimum hardware requirements are:

- Computer or Laptop
- Intel i3 / i5 Processor
- 4 GB RAM or above
- 500 GB Hard Disk
- Internet Connection
- Printer (optional)

2.8 Software Requirements

Required software tools are:

- Windows 10 / 11 Operating System
- Salesforce CRM Platform
- OCR Tool (SharinPix / OCR API)

- OpenAI / Einstein AI
- Google Chrome / Web Browser
- Microsoft Office for documentation

2.9 Feasibility Study

Economic Feasibility

The system reduces manual labor cost and increases productivity, making it cost-effective.

Technical Feasibility

Required technologies such as Salesforce, OCR, and AI are available and easy to integrate.

Operational Feasibility

The system is user-friendly and can be adopted easily by customer support teams.

2.10 Conclusion

System Requirement Analysis shows that the proposed automated solution is technically possible, economically beneficial, and operationally efficient. It can solve the major problems of manual case management and improve customer service performance.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Introduction

System Analysis is the process of studying the working of the proposed system in detail. It helps in understanding how the system receives inputs, processes data, and provides outputs. For the project **Automated Service Case Management using Salesforce, OCR & AI**, system analysis explains how customer complaints are handled automatically from beginning to end.

This chapter describes the working flow, input-output process, classification methods, and data movement inside the system.

S.No.	Case Type	Description
1	Complaint	Product issue reported
2	Invoice Inquiry	Invoice related issue
3	Quote Acceptance	Customer accepted quotation
4	Quote Rejection	Customer rejected quotation
5	Reschedule Request	Service date change request
6	Cancellation	Service cancellation request
7	Internal Notification	Internal communication

Table 3.1 Case Classification Types

```
Customer Complaint Email
        ↓
Salesforce Case Creation
        ↓
Attachment Processing
        ↓
OCR Text Extraction
        ↓
AI Prediction
        ↓
Priority Set
        ↓
Case Assigned
        ↓
Resolution
```

Fig. 3.1 System Workflow Diagram

3.2 Purpose of the System

```
Customer Email
       ↓
Configured Support Email
       ↓
Salesforce Email-to-Case
       ↓
Case Record Generated
       ↓
Agent Dashboard
```

Fig. 3.2 Email-to-Case Process flow

S.No.	Customer Input	Expected Output
1	Washing machine leakage image	Complaint - High Priority
2	Invoice PDF attached	Invoice Inquiry
3	AC service reschedule mail	Reschedule Request
4	TV repair cancel mail	Cancellation
5	Quote accepted mail	Quote Acceptance

Table 3.2 Sample Customer Complaint Inputs

3.3 Input to the System

The system receives the following inputs from customers:

- Complaint emails
- Product images
- Invoice PDF files
- Warranty documents
- Service reschedule requests
- Quote approval/rejection mails
- Internal notifications

These inputs are sent through email and processed automatically.

```
PDF / Image Attachment
        ↓
OCR Engine
        ↓
Text Detection
        ↓
Extracted Data
        ↓
Stored in Salesforce Field
```

Fig. 3.3 OCR Text Extraction Process

3.4 Process Analysis

The complete system process works as follows:

Step 1: Email Received

Customer sends an email describing the issue with attachments such as images or invoices.

Step 2: Case Creation

Salesforce Email-to-Case automatically creates a new service case.

Step 3: OCR Processing

OCR scans attached images and PDF documents to extract useful text such as invoice number, warranty date, product model, etc.

Step 4: AI Classification

Artificial Intelligence analyzes the complaint text and OCR data to identify:

- Complaint type
- Priority level
- Department required

Step 5: Case Assignment

The case is automatically assigned to the correct service team.

Step 6: Tracking and Resolution

Support staff solve the issue and update the case status.

```
Complaint Text
       ↓
AI Model Analysis
       ↓
Issue Type Detection
       ↓
Priority Prediction
       ↓
Case Category Output
```

Fig. 3.4 AI Classification Model Flow

3.5 Output of the System

The system provides the following outputs:

- Created service case ID
- Complaint category
- Priority level
- Extracted document data
- Assigned department
- Case status updates
- Reports and dashboards

CHAPTER 4: SYSTEM DESIGN

4.1 Introduction

System Design is an important phase of software development in which the structure, modules, and working architecture of the system are planned. It converts the requirements of the system into a blueprint for implementation. For the project **Automated Service Case Management using Salesforce, OCR & AI**, system design defines how customer emails, OCR processing, AI classification, and case routing are connected.

This chapter explains the architecture design, modules used, data design, and interface design of the proposed system.

```
Customer
   ↓
Email Server
   ↓
Salesforce CRM
   ├── Case Object
   ├── OCR Integration
   ├── AI Engine
   └── Routing Rules
          ↓
      Support Team
```

Fig. 4.1 System Architecture Diagram

S.No.	Module Name	Function
1	Email Module	Receive customer emails
2	Case Module	Create support case
3	OCR Module	Extract text from attachments
4	AI Module	Predict category & priority
5	Routing Module	Assign to support team
6	Dashboard Module	View reports

Table 4.1 Modules Of Proposed System

4.2 Architecture Design

The overall architecture of the system is designed in a step-by-step automated flow.

Architecture Flow:

Customer Email → Salesforce Email-to-Case → OCR Engine → AI Classification Engine → Case Assignment → Support Team → Resolution Update

Description:

- Customer sends complaint through email.
- Salesforce automatically creates a service case.
- OCR extracts text from images and PDF attachments.
- AI analyzes complaint type and urgency.
- Case is assigned to correct department.
- Support team resolves issue and updates status.

```
Salesforce
   ├── Email-to-Case
   ├── Flow Builder
   ├── Apex Trigger
   ├── OCR API
   └── AI API
```

Fig. 4.2 Salesforce Integration Architecture

4.3 Modules of the System

The system is divided into the following modules:

1. Email Handling Module

This module receives customer emails and attachments.

2. Case Creation Module

Automatically creates cases in Salesforce CRM.

3. OCR Processing Module

Reads text from invoices, warranty cards, images, and PDFs.

4. AI Classification Module

Classifies complaint type, urgency, and department.

5. Case Routing Module

Routes service requests to the appropriate support team.

6. Monitoring Module

Tracks case status, reports, and dashboards.

```
Main System
   ├── Email Module
   ├── Case Module
   ├── OCR Module
   ├── AI Module
   ├── Routing Module
   └── Reporting Module
```

Fig.4.3 Module Design Diagram

4.4 Data Design

The system stores important customer and complaint information in the database.

Main Data Fields:

- Customer Name
- Email ID
- Contact Number
- Product Name
- Product Model
- Invoice Number
- Complaint Description
- Priority Level
- Department Name
- Case Status
- Date and Time

This structured data helps in easy searching and reporting.

```
Customer ----< Case >---- Agent
                 |
                 |
            Attachments
                 |
             OCR Data
```

Fig.4.4 database/Object relationship Diagram

4.5 Interface Design

The user interface is simple and user-friendly.

Input Interfaces:

- Customer Email Inbox
- Attachment Upload Screen
- Case Entry Form
- Admin Login Panel

Output Interfaces:

- Case Dashboard
- Priority Report
- Assigned Team List
- Resolved Case Report
- Customer Status Update

4.6 Security Design

To protect customer data, the following security measures are used:

- User Login Authentication
- Role-based Access Control
- Secure Data Storage
- Authorized Case Access
- Backup and Recovery

4.7 Advantages of System Design

- Easy to understand workflow
- Faster processing
- Better modular structure
- Scalable system design
- Secure data handling
- Easy maintenance and updates

4.8 Conclusion

The system design provides a clear and efficient structure for implementing the Automated Service Case Management system. By combining Salesforce, OCR, and AI in a modular architecture, the system becomes faster, smarter, and easier to manage

CHAPTER 5: TESTING

5.1 Introduction

Testing is an important phase of software development used to verify whether the system works correctly according to requirements. It helps identify errors, improve performance, and ensure reliability of the software. For the project **Automated Service Case Management using Salesforce, OCR & AI**, testing is necessary to confirm that all modules such as email handling, OCR processing, AI classification, and case routing function properly.

This chapter explains the objectives, methods, test cases, and results of system testing.

```
Input Email
     ↓
Create Case Test
     ↓
OCR Test
     ↓
AI Prediction Test
     ↓
Routing Test
     ↓
Output Validation
```

Figure 5.1 Testing Process Flow

S.No.	Test Case	Input	Expected Result
1	Email Test	Complaint Email	Case Created
2	OCR Test	PDF Invoice	Text Extracted
3	AI Test	Complaint Text	Classified Correctly
4	Routing Test	High Priority Case	Sent to Team
5	Dashboard Test	Stored Data	Report Displayed

Table 5.1 Test Cases and Expected Results

5.2 Objectives of Testing

The main objectives of testing are:

- To verify proper working of all modules
- To identify system errors and bugs

- To ensure correct case creation from emails
- To check OCR accuracy on documents
- To validate AI classification results
- To ensure proper case routing
- To improve system reliability and performance

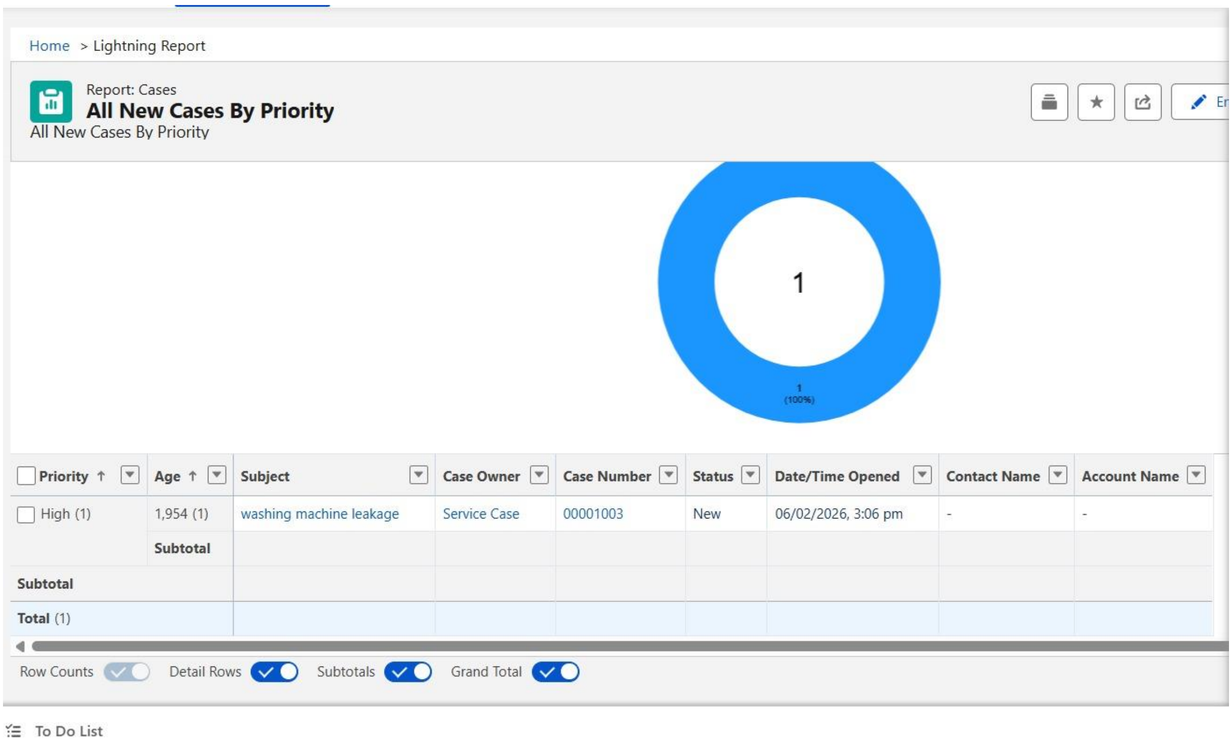

Figure 5.2 Sample Test Result Screenshot

<table>
<tr><th>Parameter</th><th>Existing System</th><th colspan="2">Proposed System</th></tr>
<tr><td>Case Creation Time</td><td>High</td><td colspan="2">Low</td></tr>
<tr><td>Accuracy</td><td>Medium</td><td colspan="2">High</td></tr>
<tr><td>Human Effort</td><td>High</td><td colspan="2">Low</td></tr>
<tr><td>Priority Detection</td><td>Manual</td><td colspan="2">Automatic</td></tr>
<tr><td colspan="2">Customer Satisfaction</td><td>Medium</td><td>High</td></tr>
</table>

Table 5.2 System Performance Comparison

5.3 Types of Testing Used

1. Unit Testing

Each module was tested individually such as email module, OCR module, and AI module.

2. Integration Testing

Different modules were tested together to ensure smooth data flow between Salesforce, OCR, and AI.

3. System Testing

The complete system was tested as a whole to verify all functionalities.

4. User Acceptance Testing

End users tested the system to confirm that it meets business requirements.

5.4 Test Cases

Test Case No.	**Input**	**Expected Output**	**Result**
TC-01	Customer complaint email	Service case created	Pass
TC-02	Invoice PDF uploaded	Text extracted by OCR	Pass
TC-03	Product damage image	OCR reads details	Pass
TC-04	Complaint text analyzed	Correct category predicted	Pass
TC-05	Urgent complaint mail	High priority assigned	Pass
TC-06	Case generated	Assigned to support team	Pass
TC-07	Case status updated	Dashboard reflects changes	Pass

5.5 Sample Test Data

Input 1:

Email Subject: Washing Machine Leakage Complaint
Attachment: Invoice PDF

Output:

- Case Created Successfully
- OCR extracted invoice details
- Priority set to High

Input 2:

Email Subject: Refrigerator Cooling Issue
Attachment: Product Image

Output:

- Case Created
- Complaint classified correctly
- Assigned to technician team

5.6 Testing Results

After testing all modules, the system performed successfully. Major observations:

- Automatic case creation worked correctly
- OCR extracted text accurately from clear images and PDFs
- AI classified cases properly
- Priority assignment was accurate
- Case routing worked smoothly
- Dashboard updated case status in real time

5.7 Errors Found and Corrections

Some minor issues were found during testing:

- Low-quality images reduced OCR accuracy
- Duplicate emails created repeated cases
- Incorrect keywords affected AI prediction

Corrections Applied:

- Improved OCR settings
- Added duplicate case detection
- Enhanced AI training data

5.8 Advantages of Testing

- Improves system quality
- Reduces chances of failure
- Increases user confidence
- Ensures smooth performance
- Provides reliable outputs

5.9 Conclusion

Testing confirmed that the Automated Service Case Management system works efficiently and accurately. All major modules passed the test cases successfully. The system is reliable, user-friendly, and ready for practical implementation.

CHAPTER 6: LIMITATIONS

6.1 Introduction

Every software system has some limitations based on technology, data quality, user behavior, and external dependencies. Although the **Automated Service Case Management using Salesforce, OCR & AI** project provides fast and intelligent customer service automation, there are certain limitations that may affect system performance in some situations.

This chapter explains the possible limitations of the proposed system.

S.No.	Limitation
1	OCR depends on image quality
2	AI requires proper training
3	Internet connection required
4	Complex handwriting may fail
5	Initial setup cost involved

Table 6.1 Limitations of Existing and Proposed System

6.2 OCR Accuracy Depends on Image Quality

OCR technology extracts text from images and PDF files. If the uploaded image is blurred, low-resolution, tilted, damaged, or unclear, the OCR engine may fail to read the content accurately.

Examples:

- Blurry invoice image
- Torn warranty card
- Handwritten unclear text
- Dark or shadowed image

This may lead to incorrect data extraction.

6.3 Dependency on Internet Connection

The system is cloud-based and uses Salesforce CRM and AI services. A stable internet connection is required for:

- Receiving emails
- Creating cases
- Running OCR services

- AI prediction
- Dashboard access

Poor internet speed may delay case processing.

6.4 AI Prediction May Not Always Be 100% Accurate

Artificial Intelligence depends on training data and keywords. If the complaint text is unclear or uses unusual language, the AI may classify the case incorrectly.

Examples:

- Short complaint messages
- Mixed language text
- Incomplete complaint details
- New issue types not in training data

Regular model updates are required.

6.5 Limited Handwritten Text Recognition

OCR works better on printed text than handwritten text. If customers upload handwritten bills or notes, extraction accuracy may be low.

6.6 Duplicate Email Cases

If customers send the same complaint multiple times, duplicate service cases may be created unless duplicate detection rules are properly configured.

6.7 Initial Setup Cost

Implementation of Salesforce CRM, OCR APIs, AI tools, and customization may require initial investment in software licenses and development.

6.8 User Training Required

Employees need basic training to use dashboards, reports, case updates, and automated workflows effectively.

6.9 Data Privacy and Security Concerns

The system handles customer personal data such as:

- Name
- Email ID
- Contact Number
- Product Details
- Invoice Data

Proper security and access control are necessary to prevent misuse.

6.10 Limited Language Support

If OCR or AI models are configured only for English, complaints in regional languages may not be processed accurately.

6.11 Conclusion

Although the system has some limitations, most of them can be reduced through better image quality, regular AI training, secure configuration, and continuous improvement. The advantages of automation are much greater than these limitations.

CHAPTER 7: FUTURE SCOPE

7.1 Introduction

Future Scope refers to the possible improvements and advanced features that can be added to the existing system in coming years. The project **Automated Service Case Management using Salesforce, OCR & AI** already provides an efficient solution for handling customer complaints automatically. However, with new technologies and growing customer expectations, the system can be further enhanced to make it smarter, faster, and more user-friendly.

This chapter explains the future enhancements that can be implemented in the system.

```
Future Scope
   ├── Chatbot Support
   ├── WhatsApp Cases
   ├── Voice Complaint
   ├── Mobile App
   └── Predictive Service Alerts
```

Figure 7.1 Future Scope Models

S.No.	Enhancement
1	Chatbot Support
2	WhatsApp Integration
3	Voice Complaint System
4	Multi-language OCR
5	Mobile Application
6	Predictive Maintenance

Table 7.1 Future Enhancements

7.2 Chatbot Integration

An AI-powered chatbot can be integrated with the system to provide instant customer support.

Benefits:

- 24/7 customer assistance
- Instant reply to common questions
- Complaint registration through chat
- Reduced workload on support staff

Customers can raise complaints directly through website chat or mobile app.

7.3 Voice-Based Complaint Registration

In future, the system can support voice input where customers can speak their complaint instead of typing emails.

Benefits:

- Easy complaint registration
- Useful for non-technical users
- Faster communication
- Better accessibility

Voice data can be converted into text using Speech Recognition technology.

7.4 WhatsApp and Social Media Integration

The system can be connected with platforms such as:

- WhatsApp
- Facebook Messenger
- Instagram
- Telegram

Customers can submit complaints through these channels, and cases can be created automatically in Salesforce.

7.5 Multi-Language Support

Future versions can support multiple languages such as:

- Hindi
- English
- Marathi
- Gujarati
- Tamil
- Telugu

This will help customers from different regions communicate easily.

7.6 Predictive Maintenance System

Using AI and machine learning, the system can predict product failures before they occur.

Example:

- Refrigerator cooling issue prediction
- Washing machine motor warning
- AC gas leakage alert

This helps in preventive maintenance and better customer service.

7.7 Mobile Application Development

A mobile app can be developed for both customers and service staff.

Features for Customers:

- Register complaint
- Upload images
- Track case status
- Chat support

Features for Technicians:

- View assigned cases
- Update status
- Customer location access
- Job completion report

7.8 Advanced Analytics Dashboard

Future dashboards can provide:

- Complaint trends
- Product-wise issue reports
- Technician performance reports
- Resolution time analysis

- Customer satisfaction scores

This helps management in decision making.

7.9 Smart Auto Reply System

AI can automatically send smart replies to customers such as:

- Complaint received confirmation
- Estimated resolution time
- Technician visit details
- Case closed notification

This improves communication and trust.

7.10 IoT Device Integration

Smart appliances connected through IoT can directly send fault alerts to the system.

Example:

- Smart AC sends gas leakage alert
- Washing machine sends motor fault signal
- Refrigerator sends cooling failure alert

Cases can be created automatically without customer complaint.

7.11 Conclusion

The future scope of this project is very wide. By integrating chatbot support, mobile apps, predictive AI, IoT devices, and multi-language communication, the system can become a complete smart customer service platform. These improvements will make service management faster, more intelligent, and customer-friendly.

CHAPTER 8: CONCLUSION

8.1 Introduction

Conclusion is the final chapter of the project report that summarizes the overall work, achievements, and usefulness of the system. The project **Automated Service Case Management using Salesforce, OCR & AI** was developed to improve customer support operations by automating complaint handling and service request management.

This project successfully combines modern technologies such as Salesforce CRM, Optical Character Recognition (OCR), and Artificial Intelligence (AI) to create a smart and efficient customer service solution.

```
Dashboard
 ├── Total Cases
 ├── Open Cases
 ├── High Priority Cases
 ├── Closed Cases
 └── Avg Resolution Time
```

Table 8.1 Final Output Dashboard

S.No.	Outcome
1	Faster case handling
2	Reduced manual effort
3	Improved accuracy
4	Better customer support
5	Smart automation
6	Increased efficiency

Table 8.1 Final Project Outcomes

8.2 Summary of the Project

In many organizations, customer complaints are handled manually through emails and support staff. This traditional process is slow, error-prone, and difficult to manage when the number of requests is high.

To solve this problem, the proposed system automates the complete workflow:

- Customer emails are converted into service cases automatically
- OCR extracts useful data from invoices, images, and PDF files

- AI classifies complaint type and priority level
- Cases are routed to the correct support team
- Case status can be tracked easily through dashboards

This reduces manual effort and improves efficiency.

8.3 Major Achievements

The project achieved the following objectives successfully:

- Automated service case creation
- Fast processing of customer complaints
- Improved accuracy in classification
- Reduced response time
- Better complaint tracking
- Reduced workload on employees
- Enhanced customer satisfaction

8.4 Benefits of the System

The system provides many benefits to organizations:

- Saves time and operational cost
- Reduces human errors
- Handles large volume of requests
- Provides smart decision making using AI
- Improves communication with customers
- Increases overall productivity

8.5 Practical Importance

This project is highly useful for:

- Home appliance companies
- Electronics service centers

- CRM support teams
- Warranty departments
- Complaint management organizations

It can also be adopted by any business that receives customer support requests regularly.

8.6 Final Result

After testing and analysis, the system performed successfully. Salesforce automation, OCR extraction, and AI prediction worked effectively together. The system is reliable, scalable, and suitable for real-world implementation.

8.7 Closing Statement

The **Automated Service Case Management using Salesforce, OCR & AI** project proves that intelligent automation can transform traditional customer support systems into faster, smarter, and more efficient service platforms. It is a modern solution that helps businesses deliver better service quality and customer satisfaction.

8.8 Conclusion Statement

Hence, the project is completed successfully and meets all the required objectives of automated customer service case management.

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CHAPTER 9

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